

Bitcoin Residual Dynamics Playground

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$$\dot{y}_1 = y_2 - Z_A y_2 z + Z_{\text{mix}} (y_1 - y_2) \quad (1)$$

$$\dot{y}_2 = \frac{\alpha y_2 |y_2|^{M-1} - \gamma y_1 |y_1|^{N-1}}{t} + Z_B y_1 z - \frac{\kappa \operatorname{sgn}[X(t)] + y_2}{t} + \frac{\sigma(t)}{t} \dot{W}_t \quad (2)$$

$$\dot{z} = -Z_C z + Z_D y_1 + Z_E y_1 y_2 \quad (3)$$

The z -terms act only for $t \geq Z_{\text{start}}$; before, $z \equiv 0$.

1. LPPL Oscillator (Core)

$$\dot{y}_1 = y_2 \quad (4)$$

$$\dot{y}_2 = \alpha y_2 |y_2|^{M-1} - \gamma y_1 |y_1|^{N-1} \quad (5)$$

2. Diffusion Extension

$$\dot{y}_2 = -\frac{\kappa \operatorname{sgn}[X(t)] + y_2}{t} + \frac{\sigma(t)}{t} \dot{W}_t \quad (6)$$

$$X(t) = n(t) - \mu(t), \quad n(t) := n_{PL} + t y_2 \quad (7)$$

For the sign-diffusion part alone and a fixed target μ , the exponent $n = n_{PL} + t y_2$ obeys $dn = -\kappa \operatorname{sgn}[X] dt + \sigma dW_t$, whose stationary distribution is $\text{Laplace}(\mu, b)$ with $b = \sigma^2/(2\kappa)$. The noise is therefore set by the measured Laplace scale b_c of the daily exponent per cycle, $\sigma = \sqrt{2\kappa b_c}$. The full system, with spring, z and a time-dependent μ , has no stationary Laplace law.

Target exponent $\mu(t)$. In the simplest case $\mu(t)$ is piecewise constant per halving cycle c and phase,

$$\mu(t) = n_{PL} + \Delta_{c,P}, \quad P \in \{P_1, P_2, P_3\},$$

with $P_1 : 100 \leq d_c \leq 550$, $P_2 : 550 < d_c \leq 950$ and P_3 otherwise, where d_c is the position within the halving cycle, rescaled from the actual cycle length to a reference length of 1392.5 d. The offsets $\Delta_{c,P}$ are the measured mean deviations of the daily exponent from n_{PL} . At phase boundaries μ jumps, so X jumps while n evolves continuously. Alternatively, $\mu(t)$ can follow a centred moving average of the measured exponent (window 30–90 d).

3. Wang-like 2-Scroll Extension

$$\dot{y}_1 = y_2 - Z_A y_2 z + Z_{\text{mix}} (y_1 - y_2) \quad (8)$$

$$\dot{y}_2 = \alpha y_2 |y_2|^{M-1} - \gamma y_1 |y_1|^{N-1} + Z_B y_1 z \quad (9)$$

$$\dot{z} = -Z_C z + Z_D y_1 + Z_E y_1 y_2 \quad (10)$$

$$\dot{y}_1 = Z_{\text{mix}}(y_1 - y_2) - y_2 z \quad (11)$$

$$\dot{y}_2 = -b y_2 + y_1 z \quad (12)$$

$$\dot{z} = -Z_C z + Z_D y_1 + y_1 y_2 \quad (13)$$

LPPL Oscillator

y_1	price residual in log-space (displacement from power-law)
y_2	velocity variable, $y_2 = \dot{y}_1$ without z -coupling
α, γ, M, N	LPPL parameters

Laplace Diffusion

n_{PL}	Bitcoin power-law mean exponent
$n(t)$	exponent variable: $n_{PL} + t y_2$
$\mu(t)$	target exponent, time-dependent (phases P_1 – P_3 or moving average)
$X(t)$	deviation from the target: $n(t) - \mu(t)$
κ	restoring drift (sign-based), per day
b_c	Laplace scale of the daily exponent per cycle, $\sigma = \sqrt{2\kappa b_c}$
$\sigma(t)$	noise diffusion coefficient
$\dot{W}_t$	Gaussian white noise increment

Wang 2-Scroll

z	feedback memory
Z_A, Z_B	active coupling gains
Z_C, Z_D, Z_E	feedback dynamics
Z_{mix}	mixing term
Z_{start}	day from which z is active
